US012404895B2

(12) **United States Patent**
Shinoda et al.

(10) **Patent No.: US 12,404,895 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **FOIL BEARING**

(71) Applicant: **KABUSHIKI KAISHA TOYOTA JIDOSHOKKI**, Kariya (JP)

(72) Inventors: **Fumiya Shinoda**, Kariya (JP); **Fumihiro Suzuki**, Kariya (JP); **Takahito Kunieda**, Kariya (JP); **Kenta Nakane**, Kariya (JP)

(73) Assignee: **KABUSHIKI KAISHA TOYOTA JIDOSHOKKI**, Kariya (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 126 days.

(21) Appl. No.: **18/472,465**

(22) Filed: **Sep. 22, 2023**

(65) **Prior Publication Data**

US 2024/0125349 A1 Apr. 18, 2024

(30) **Foreign Application Priority Data**

Oct. 13, 2022 (JP) 2022-164877
May 19, 2023 (JP) 2023-082789

(51) **Int. Cl.**
F16C 17/02 (2006.01)

(52) **U.S. Cl.**
CPC ***F16C 17/024*** (2013.01)

(58) **Field of Classification Search**
CPC F16C 17/024
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,560,922 B2 * 1/2023 Aguilar F16C 17/024
11,739,761 B2 * 8/2023 Okano F16C 17/024
415/103
2022/0099102 A1 3/2022 Okano et al.

FOREIGN PATENT DOCUMENTS

JP 2022-057207 A 4/2022

* cited by examiner

Primary Examiner — James Pilkington
(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC

(57) **ABSTRACT**

A foil bearing includes a bearing housing, a top foil, a bump foil, and two retaining portions. A holding groove is formed in the inner circumferential surface of the bearing housing to extend in an axial direction. The top foil and the bump foil each include an insertion plate portion, which is inserted into the holding groove and extends in the axial direction. Each of the retaining portions includes a facing surface that faces the insertion plate portion in the axial direction. In the axial direction, the distance between the facing surfaces of the two retaining portions is longer than the length of the insertion plate portion. At least one of the insertion plate portions includes a bent plate portion that is bent to contact the facing surface.

9 Claims, 8 Drawing Sheets

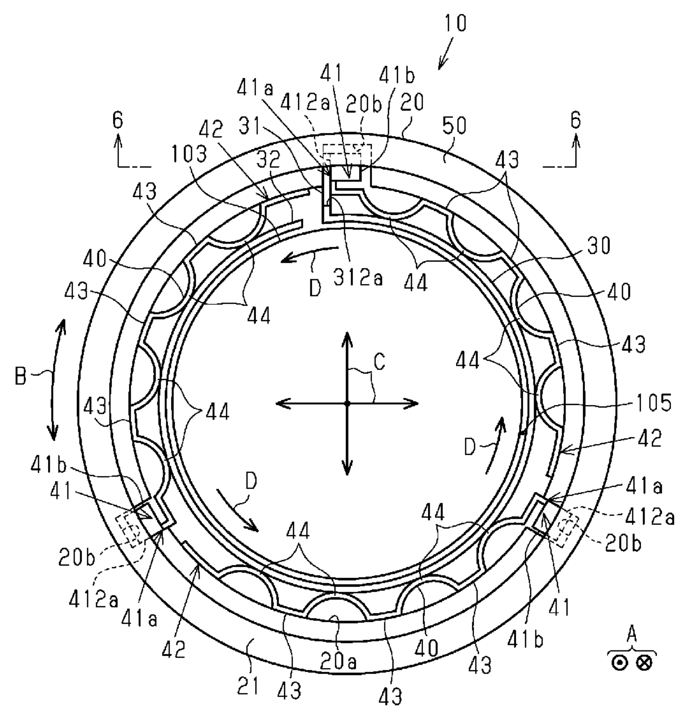


Fig.1

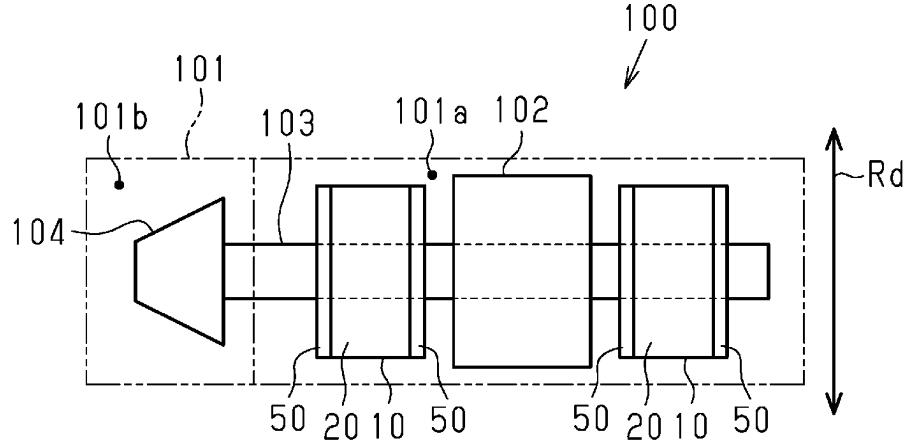
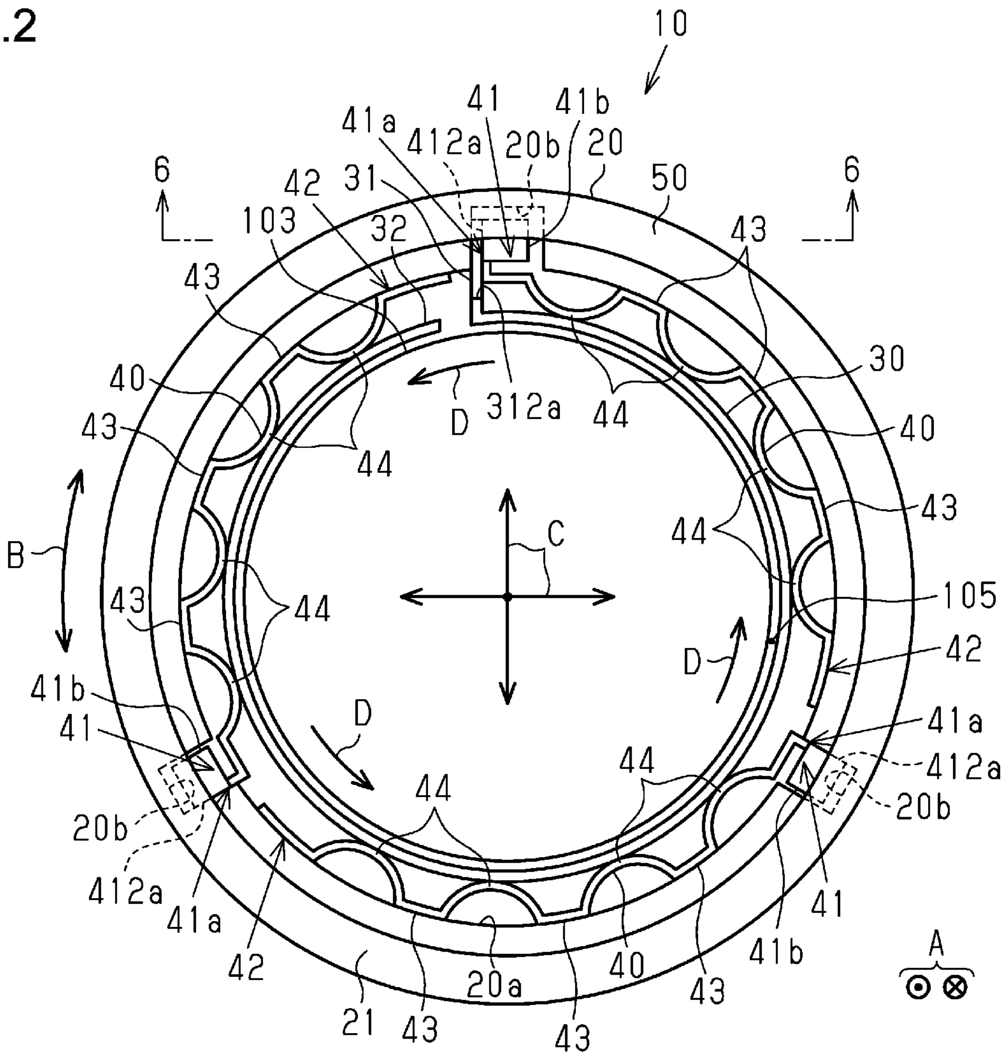

100
101
101b
103
101a
102
Rd
104
50
20
10
50
50
20
10
50

Fig.2


10
41
41b
41a
412a
20b
20
6
42
31
50
6
103
32
43
43
43
30
40
D
312a
44
40
43
44
43
44
B
C
105
43
44
D
42
41b
41a
41
D
412a
20b
44
20b
44
41
412a
41a
40
41b
42
20a
43
A
21
43
43

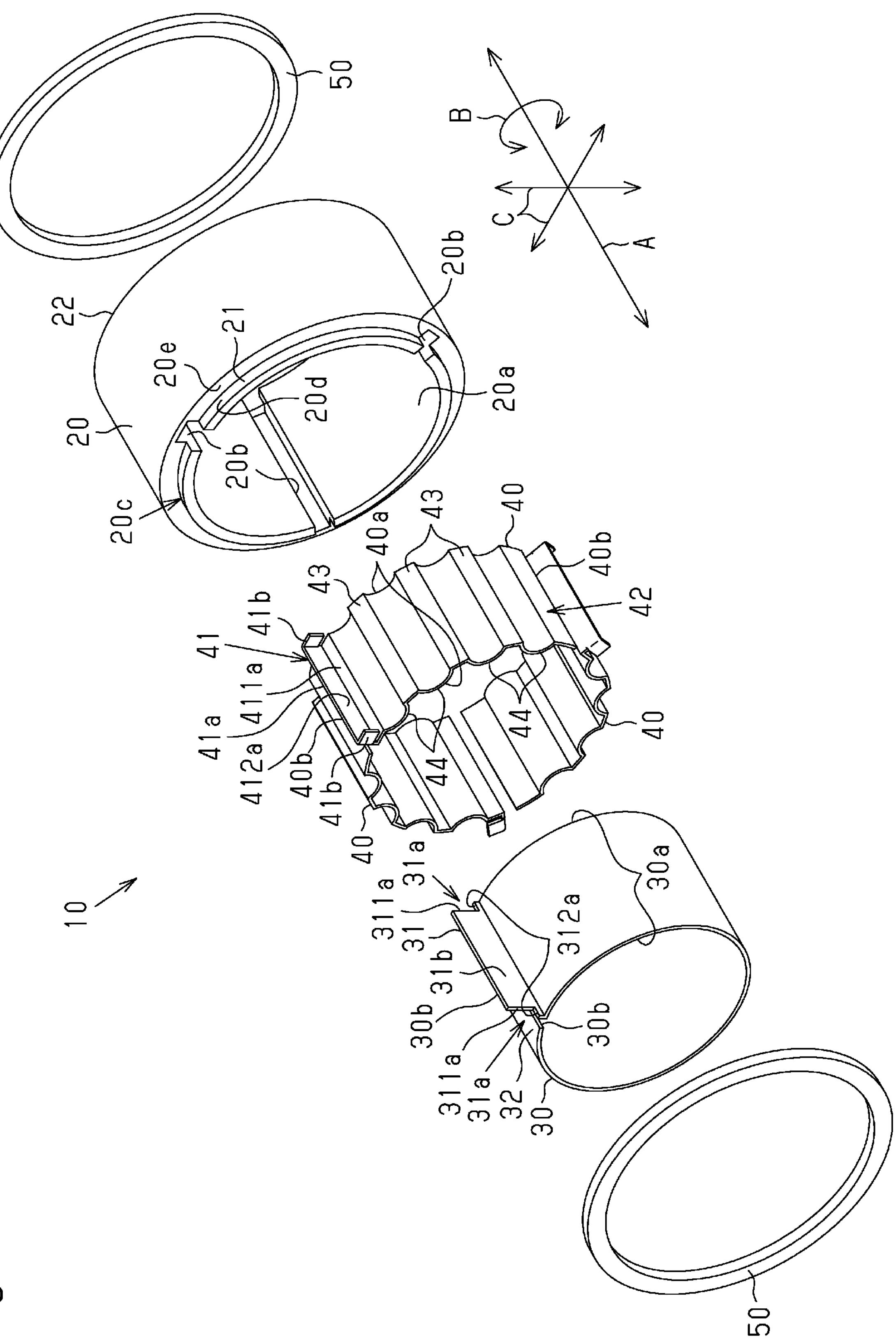

50
B
C
A
22
21
20b
20e
20d
20a
20
20b
20c
40
43
40a
43
40b
42
41b
41
411a
41a
412a
40b
41b
40
44
44
40
10
31a
311a
31
31b
30b
311a
31a
32
30
312a
30a
30b
50

Fig.3

Fig.4

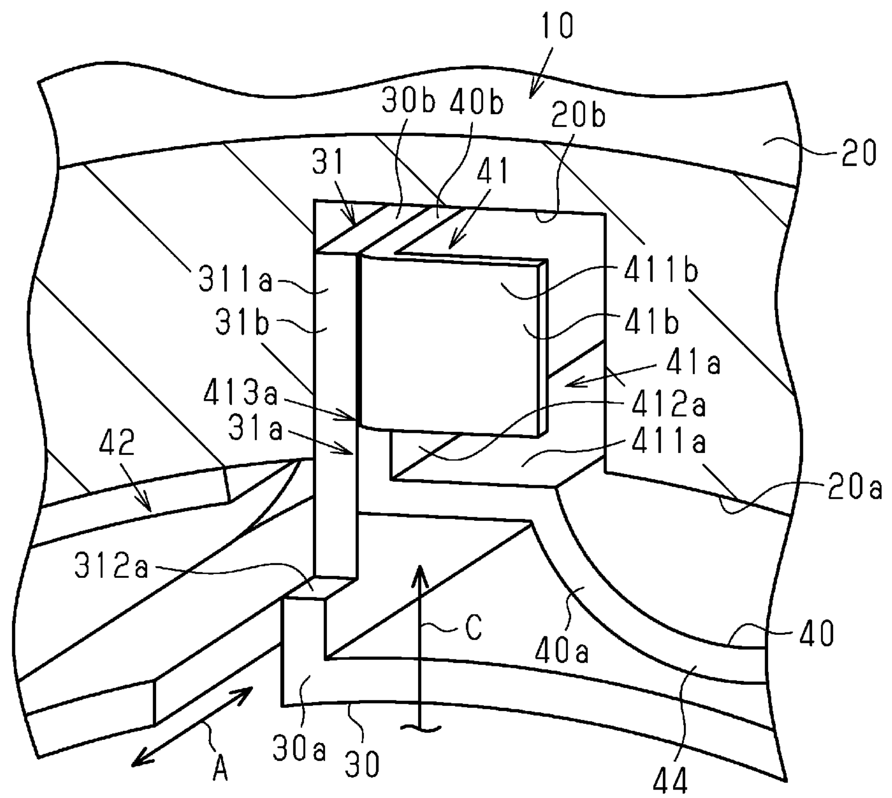

10
30b 40b
20b
20
31
41
311a
411b
31b
41b
41a
413a
412a
31a
411a
42
20a
312a
C
40
40a
A
30a
30
44

Fig.5

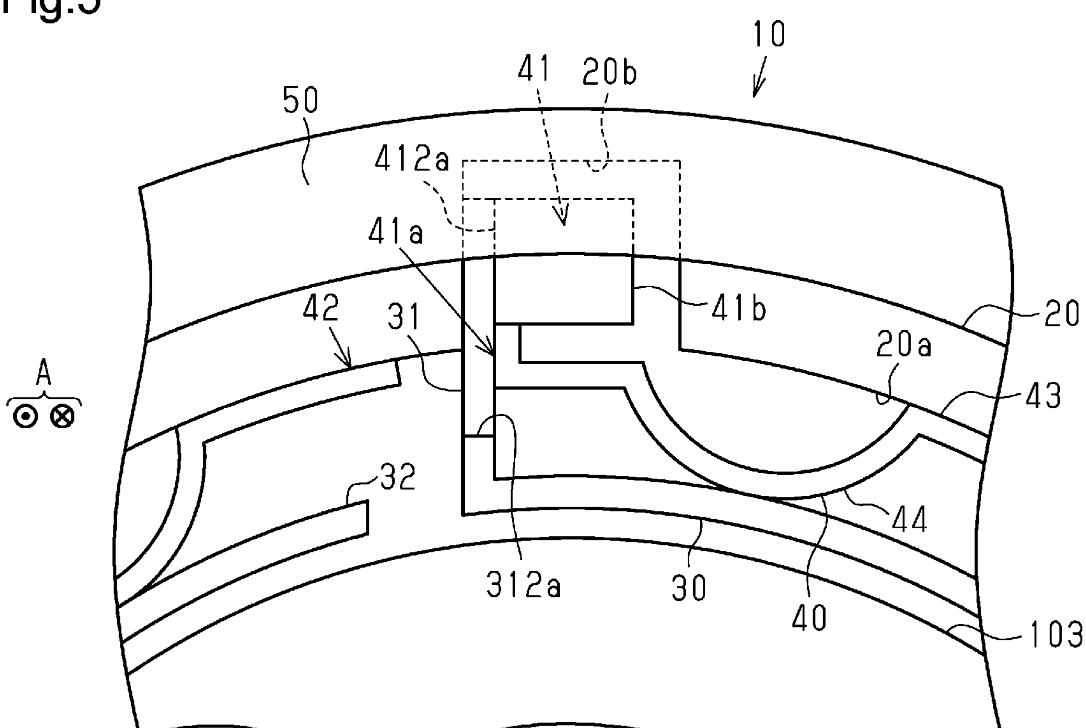
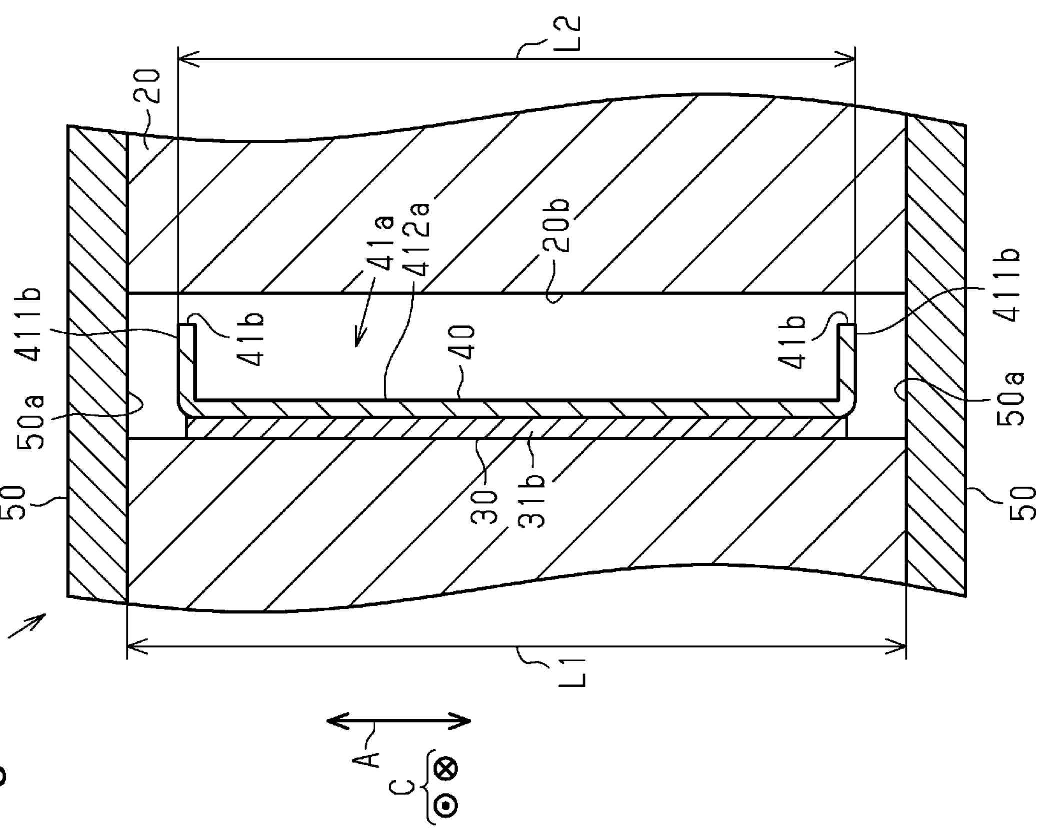
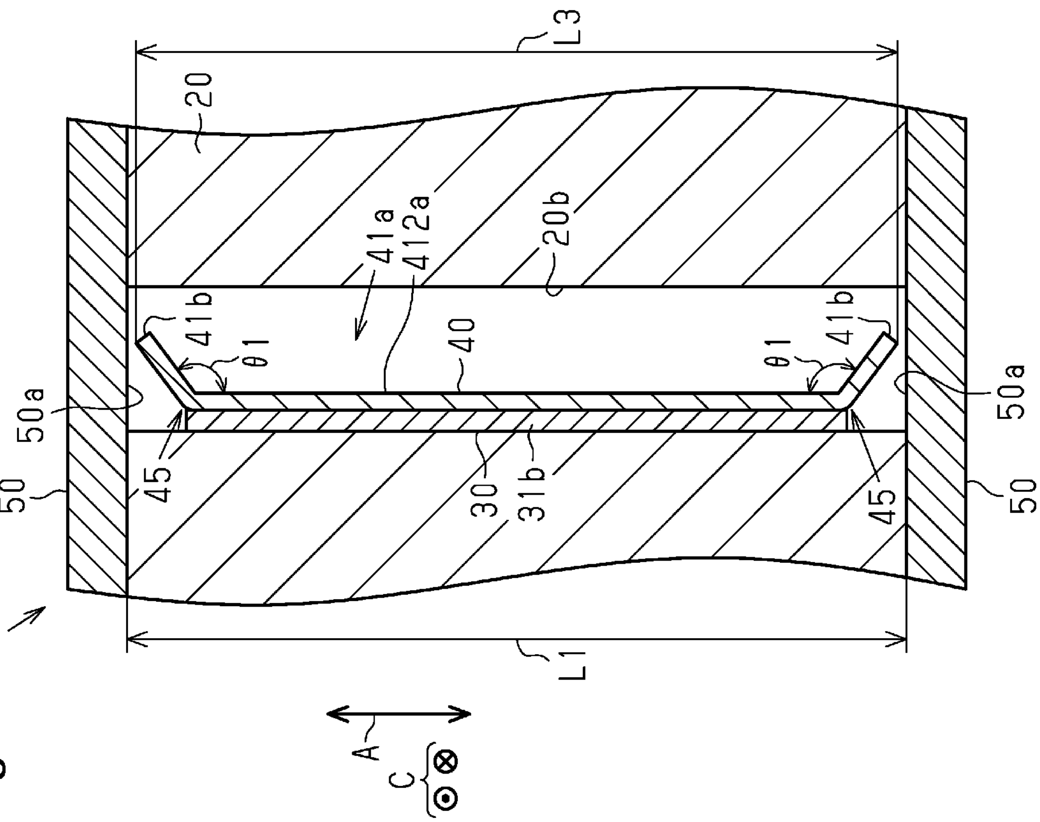

10
41
20b
50
412a
41a
41b
42
31
20
20a
43
A
32
44
312a
30
40
103

Fig.6

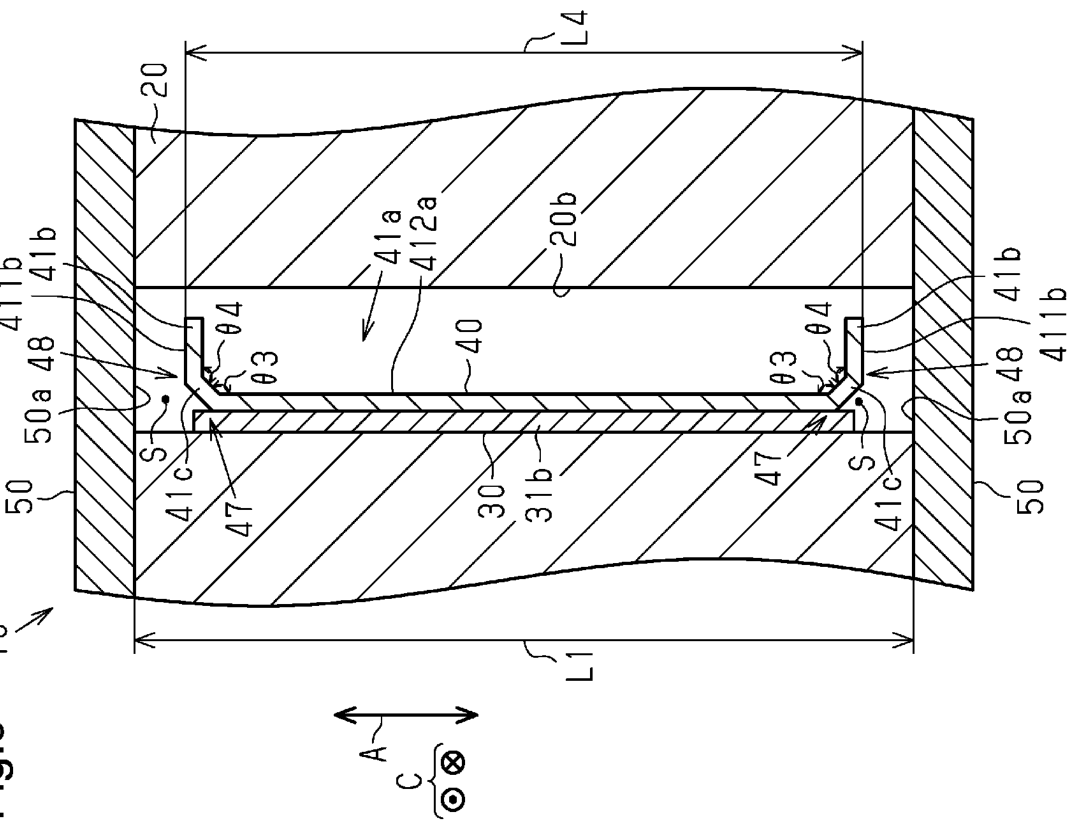


L2
20
20b
41a
412a
411b
41b
40
41b
411b
50a
50a
50
30
31b
50
10
L1
A
C

Fig.7

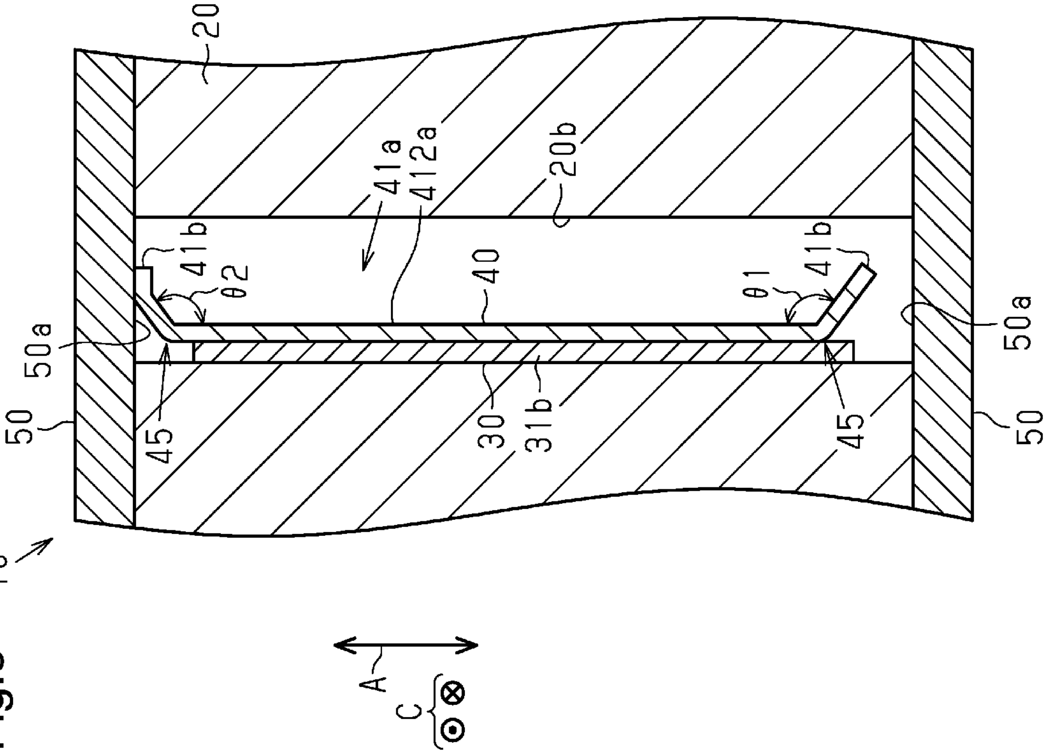


L3
20
20b
41a
412a
41b
41b
40
θ1
θ1
50a
50a
45
45
50
30
31b
50
10
L1
A
C



L4
20
20b
41a
412a
41b
411b
θ4
θ3
48
50a
40
S
41c
47
30
31b
50
10
L1
A
C

Fig.9



20
20b
41a
412a
41b
θ2
θ1
40
50a
50
45
30
31b
10
A
C

Fig.8

Fig.11

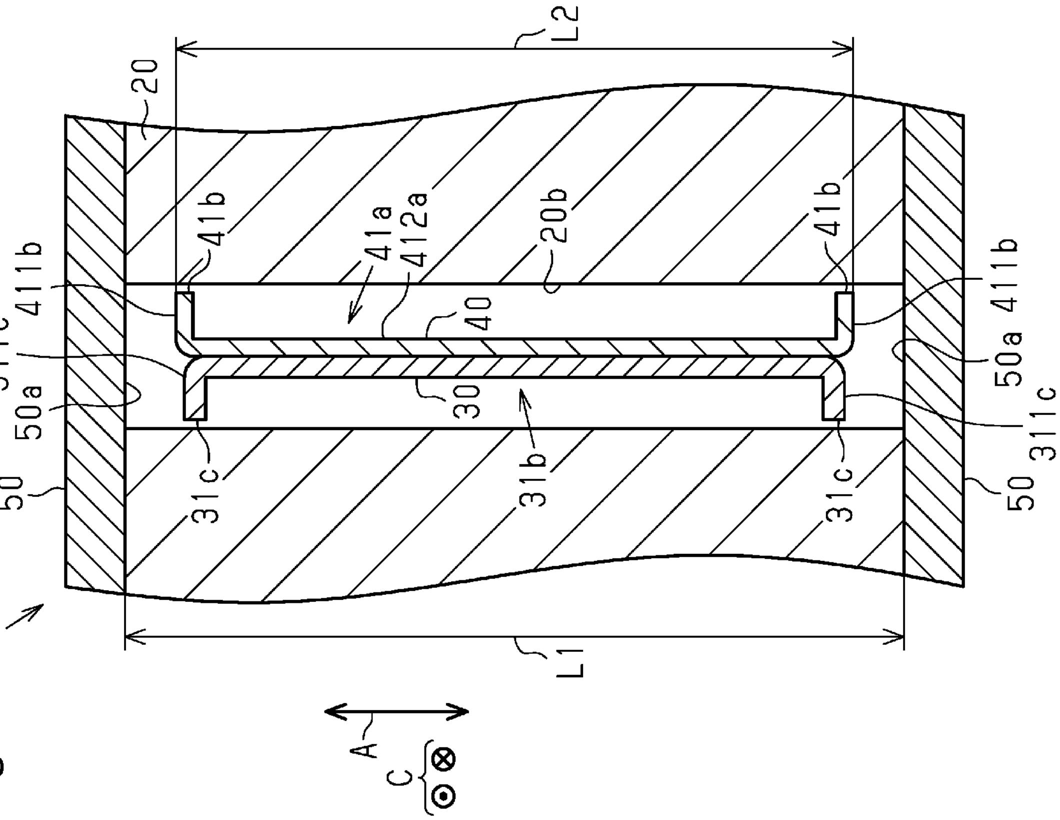
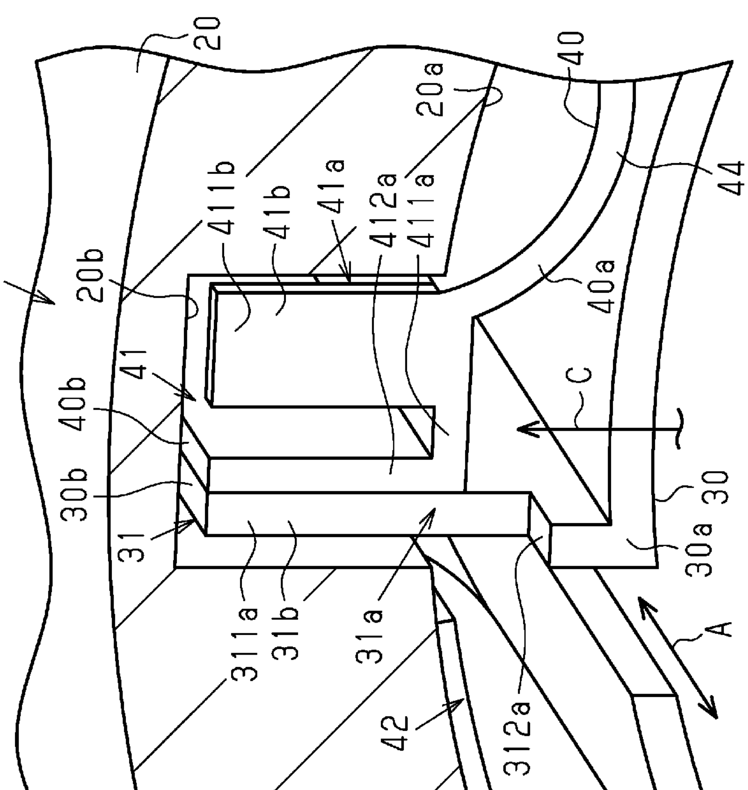

10
20
50
50a
311c
411b
41b
41a
412a
20b
40
30
31c
31b
L1
L2
A
C

Fig.10


10
20
20a
20b
40
44
40a
40b
30
30a
30b
41
41a
41b
411a
411b
412a
31
31a
31b
311a
312a
42
C
A

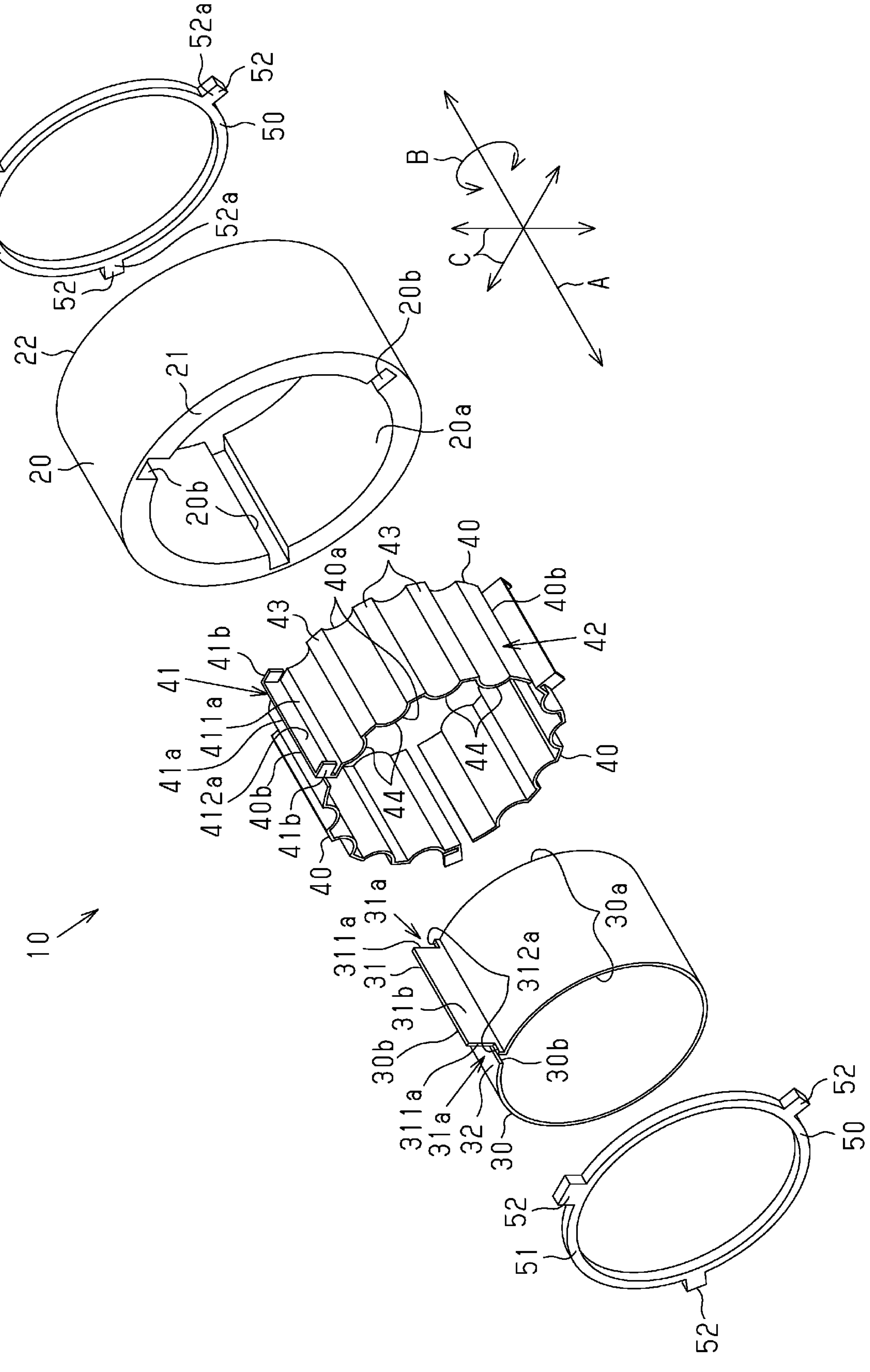

52a
52
50
52
52a
51
52a
52
B
C
A
22
21
20b
20
20b
20a
40
43
40a
43
40b
41b
41
42
411a
41a
412a
40b
41b
44
44
40
40
10
31a
311a
31
31b
30b
311a
31a
32
30
30a
312a
30b
52
50
52
51
52

Fig.12

Fig.14

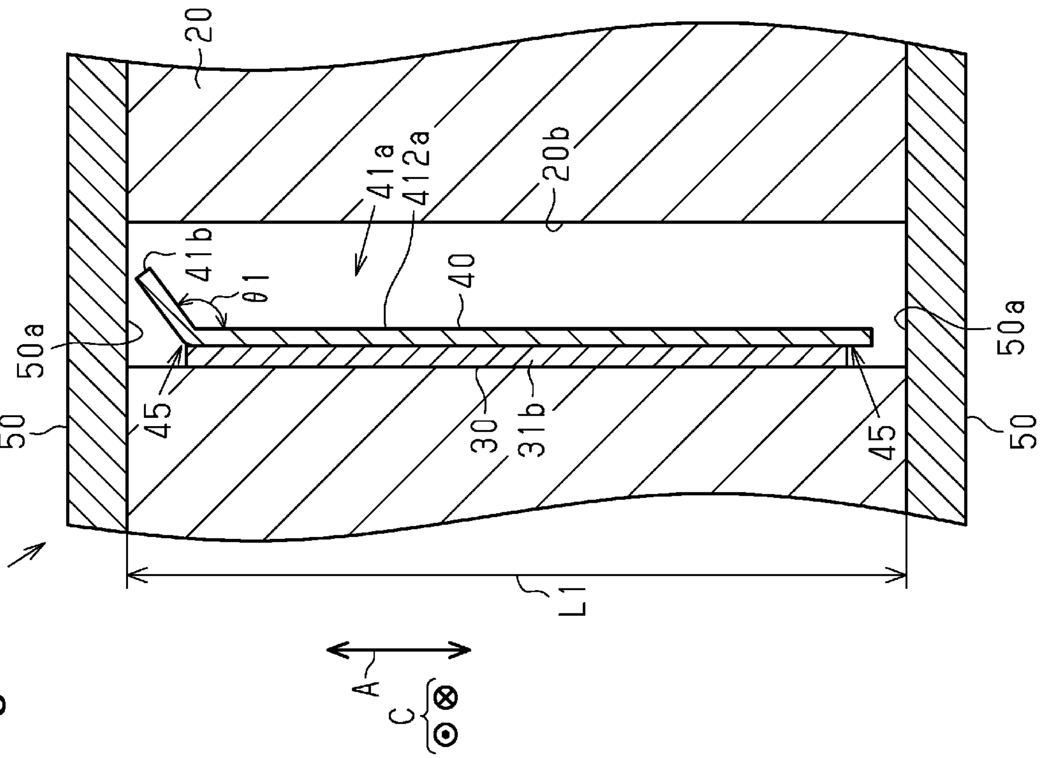
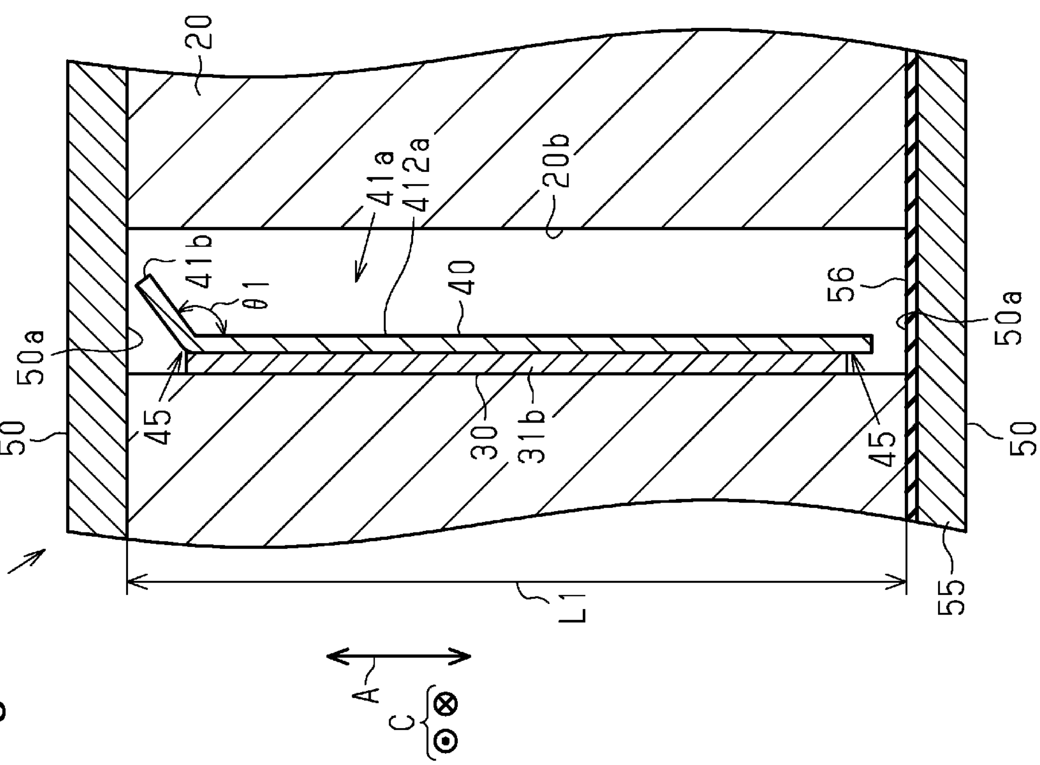

20
20b
41a
412a
41b
θ1
40
50a
50a
50
50
45
45
30
31b
10
L1
A
C

Fig.13


20
20b
41a
412a
41b
θ1
40
56
50a
50a
50
50
45
45
30
31b
55
10
L1
A
C

FOIL BEARING

BACKGROUND

1. Field

The present disclosure relates to a foil bearing that supports a rotary shaft in a radial direction.

2. Description of Related Art

Japanese Laid-Open Patent Publication No. 2022-57207 discloses a centrifugal compressor equipped with a foil bearing that supports a rotary shaft in a radial direction.

This foil bearing includes a tubular bearing housing, a top foil, a bump foil, and a retaining portion. A rotary shaft is inserted into the bearing housing. The top foil is disposed between the rotary shaft and the bearing housing. The bump foil is disposed between the bearing housing and the top foil. The bump foil elastically supports the top foil. A holding groove is formed in the inner circumferential surface of the bearing housing. The holding groove opens at one end in the axial direction of the bearing housing. A part of the top foil and a part of the bump foil are inserted into the holding groove. The retaining portion faces the top foil and the bump foil in the axial direction of the bearing housing. Even if the top foil and the bump foil move in the axial direction of the bearing housing, the top foil and the bump foil contact the retaining portion. The retaining portion prevents the top foil and the bump foil from being dislodged from the bearing housing.

When at least one of the top foil or the bump foil contacts the retaining portion, at least one of the top foil or the bump foil may be pressed against the retaining portion more than necessary. This may cause the retaining portion and at least one of the top foil or the bump foil to wear.

SUMMARY

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

In one general aspect, a foil bearing supports a rotary shaft in a radial direction. The foil bearing includes a tubular bearing housing into which the rotary shaft is inserted, a thin plate-shaped top foil disposed between the rotary shaft and the bearing housing, a thin plate-shaped bump foil that is disposed between the bearing housing and the top foil to elastically support the top foil, and two retaining portions facing the top foil and the bump foil in an axial direction of the bearing housing and preventing the top foil and the bump foil from being dislodged from the bearing housing. A holding groove is formed in an inner circumferential surface of the bearing housing to extend in the axial direction. The top foil and the bump foil each include an insertion plate portion, which is inserted into the holding groove and extends in the axial direction. Each of the retaining portions includes a facing surface that faces the insertion plate portion in the axial direction. In the axial direction, a distance between the facing surfaces of the two retaining portions is longer than a length of the insertion plate portion. At least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil includes a bent plate portion at at least one of opposite ends in the axial direction. The bent plate portion is bent to contact the corresponding facing surface.

Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic diagram showing a centrifugal compressor equipped with a foil bearing according to a first embodiment.

FIG. **2** is a front view of the foil bearing shown in FIG. **1**.

FIG. **3** is an exploded perspective view of the foil bearing shown in FIG. **2**.

FIG. **4** is a perspective view of a bent plate portion of the foil bearing shown in FIG. **2**.

FIG. **5** is a diagram showing a positional relationship between a retaining member and the bent plate portion in the foil bearing shown in FIG. **2**.

FIG. **6** is a cross-sectional view taken along line **6**-**6** of FIG. **2**.

FIG. **7** is a cross-sectional view of a foil bearing according to a second embodiment.

FIG. **8** is a cross-sectional view showing operation of the foil bearing shown in FIG. **7**.

FIG. **9** is a cross-sectional view of a foil bearing according to a third embodiment.

FIG. **10** is a perspective view of a bent plate portion in a foil bearing according to a modification.

FIG. **11** is a cross-sectional view of a foil bearing according to a modification.

FIG. **12** is an exploded perspective view of a foil bearing according to a modification.

FIG. **13** is a cross-sectional view of a foil bearing according to a modification.

FIG. **14** is a cross-sectional view of a foil bearing according to a modification.

Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, except for operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

In this specification, "at least one of A and B" should be understood to mean "only A, only B, or both A and B."

First Embodiment

Foil bearings **10** according to a first embodiment will now be described with reference to FIGS. **1** to **6**. The foil bearings **10** of the present embodiment are mounted on a centrifugal compressor **100**.

Centrifugal Compressor

As shown in FIG. **1**, the centrifugal compressor **100** includes two foil bearings **10**, a housing **101**, an electric motor **102**, a rotary shaft **103**, and an impeller **104**. The electric motor **102**, the rotary shaft **103**, and the impeller **104** are accommodated in the housing **101**. The electric motor **102** is accommodated in a motor chamber **101***a* formed in the housing **101**. The impeller **104** is accommodated in an impeller chamber **101***b* formed in the housing **101**. The rotary shaft **103** extends from the motor chamber **101***a* to the impeller chamber **101***b*. The electric motor **102** is attached to the rotary shaft **103**. The impeller **104** is attached to an end of the rotary shaft **103**. When the electric motor **102** operates, the rotary shaft **103** rotates. When the rotary shaft **103** rotates, the impeller **104** rotates. When the impeller **104** rotates, fluid is drawn into the impeller chamber **101***b* from the outside of the housing **101** and is then compressed in the impeller chamber **101***b*. The fluid compressed in the impeller chamber **101***b* is discharged to the outside of the housing **101**.

The two foil bearings **10** support the rotary shaft **103** in a radial direction Rd. The two foil bearings **10** are located at the opposite sides of the electric motor **102** in the axial direction of the rotary shaft **103**. The two foil bearings **10** are fixed to the housing **101**.

Foil Bearing

As shown in FIGS. **2** and **3**, the foil bearing **10** includes a tubular bearing housing **20**, a thin plate-shaped top foil **30**, three thin plate-shaped bump foils **40**, and two retaining members **50**. The bearing housing **20** is made of, for example, aluminum.

Bearing Housing

As shown in FIG. **2**, the rotary shaft **103** is inserted into the bearing housing **20**. In the following description, the axial direction of the bearing housing **20** will simply be referred to as an axial direction A, a circumferential direction of the bearing housing **20** will simply be referred to as a circumferential direction B, and a radial direction of the bearing housing **20** will simply be referred to as a radial direction C. The bearing housing **20** includes an inner circumferential surface **20***a*. The inner circumferential surface **20***a* is a cylindrical surface that is curved in the circumferential direction B.

As shown in FIG. **3**, the bearing housing **20** includes a first end portion **21** and a second end portion **22**. The first end portion **21** and the second end portion **22** are end portions of the bearing housing **20** in the axial direction A. The first end portion **21** is one end portion of the bearing housing **20** in the axial direction A. The second end portion **22** is another end portion of the bearing housing **20** in the axial direction A. Holding grooves **20***b* are formed in the inner circumferential surface **20***a* of the bearing housing **20**. The number of the holding grooves **20***b* in the inner circumferential surface **20***a* is three. The three holding grooves **20***b* are disposed at specified intervals in the circumferential direction of the B. The holding grooves **20***b* extend in the axial direction A. Each holding groove **20***b* extends over the entire length of the inner circumferential surface **20***a* in the axial direction A.

The first end portion **21** and the second end portion **22** each include an annular cutout groove **20***c*. The cutout grooves **20***c* open outward in the radial direction C. The cutout groove **20***c* of the first end portion **21** is open toward the side opposite to the second end portion **22** in the axial direction A. The cutout groove **20***c* of the second end portion **22** is open toward the side opposite to the first end portion **21** in the axial direction A. Each cutout groove **20***c* includes a cylindrical surface **20***d*, which extends in the axial direction A, and an annular surface **20***e*, which extends in the radial direction C. The holding grooves **20***b* open in the cylindrical surface **20***d* and the annular surface **20***e*. FIG. **2** does not show the cutout groove **20***c* formed in the second end portion **22**.

Top Foil

As shown in FIG. **2**, the top foil **30** is disposed inward of the bearing housing **20**. The top foil **30** is disposed between the rotary shaft **103** and the bearing housing **20**.

As shown in FIG. **3**, the top foil **30** is formed by bending a flexible elongated metal plate into a tubular shape. The top foil **30** includes two long edges **30***a* and two short edges **30***b*. One of the two long edges **30***a* may be referred to as a first long edge **30***a*, and the other may be referred to as a second long edge **30***a*. One of the two short edges **30***b* may be referred to as a first short edge **30***b*, and the other may be referred to as a second short edge **30***b*. The top foil **30** is substantially cylindrical. The axial direction of the top foil **30** agrees with the axial direction A of the bearing housing **20**. The circumferential direction of the top foil **30** agrees with the circumferential direction B of the bearing housing **20**. The radial direction of the top foil **30** agrees with the radial direction C of the bearing housing **20**. The top foil **30** is formed by being bent into a tubular shape such that the two long edges **30***a* extend in the circumferential direction B, and the two short edges **30***b* extend in the axial direction A. The length between the long edges **30***a* is equal to the length in the axial direction A of the bearing housing **20**. The metal plate forming the top foil **30** is made of, for example, stainless steel or an Inconel® type nickel alloy.

The top foil **30** includes a fixed end **31** and a free end **32**. The fixed end **31** is formed by bending one end in the long-side direction of the metal plate forming the top foil **30** outward in the radial direction of the top foil **30**. The fixed end **31** includes the first short edge **30***b* of the top foil **30**. The free end **32** is the other end in the long-side direction of the metal plate forming the top foil **30**. The free end **32** includes the second short edge **30***b* of the top foil **30**. In the circumferential direction B, the free end **32** is spaced apart from the proximal end of the fixed end **31** and faces the proximal end of the fixed end **31**. Thus, the top foil **30** is an annular body from which a part in the circumferential direction is cut out.

The fixed end **31** includes cutout portions **31***a* and an insertion plate portion **31***b*. The cutout portions **31***a* are formed by cutting out parts of the opposite ends of the fixed end **31** in the axial direction A. One of the cutout portions **31***a* is formed by cutting out a part of the first long edge **30***a* and a part of the first short edge **30***b*. The other cutout portion **31***a* is formed by cutting out a part of the second long edge **30***a* and a part of the first short edge **30***b*. Each cutout portion **31***a* includes a first cutout edge **311***a* and a second cutout edge **312***a*. The first cutout edge **311***a* extends in the radial direction C. The second cutout edge **312***a* is continuous with the first cutout edge **311***a*. The second cutout edge **312***a* extends in the axial direction A. The first cutout edges **311***a* of the cutout portions **31***a* form the opposite ends in the axial direction A of the insertion plate portion **31***b*. The insertion plate portion **31***b* includes the first short edge **30***b*, which includes the cutout portions **31***a*.

As shown in FIG. **2**, the insertion plate portion **31***b* of the top foil **30** is inserted in one of the holding grooves **20***b*. The top foil **30** includes the insertion plate portion **31***b*, which is inserted into the holding groove **20***b* and extends in the axial direction A.

Bump Foil

The three bump foils **40** are disposed between the bearing housing **20** and the top foil **30**. The three bump foils **40** are arranged at specified intervals in the circumferential direction B. The thickness of each bump foil **40** is less than the thickness of the top foil **30**.

As shown in FIG. **3**, each bump foil **40** is formed by bending a flexible elongated metal plate into a substantially arcuate shape. Each bump foil **40** includes two long edges **40***a* and two short edges **40***b*. The short-side direction of each bump foil **40** agrees with the axial direction A of the bearing housing **20**. The length between the two long edges **40***a* is shorter than the length in the axial direction A of the bearing housing **20**. The length between the two long edges **40***a* is shorter than the length between the two long edges **30***a* of the top foil **30**. The metal plate forming each bump foil **40** is made of, for example, stainless steel or an Inconel® type nickel alloy.

Each bump foil **40** includes a fixed end **41** and a free end **42**. The fixed end **41** is an end in the long-side direction of the metal plate forming the bump foil **40**. The fixed end **41** includes one of the short edges **40***b* of the bump foil **40**. The free end **42** is the other end in the long-side direction of the metal plate forming the bump foil **40**. The free end **42** includes the other short edge **40***b* of the bump foil **40**.

As shown in FIG. **2**, each bump foil **40** has multiple valleys **43**, which contact the inner circumferential surface **20***a* of the bearing housing **20**. The valleys **43** extend along the inner circumferential surface **20***a* of the bearing housing **20**. Each bump foil **40** has multiple peaks **44**, which contact the outer circumferential surface of the top foil **30**. The peaks **44** project away from the inner circumferential surface **20***a* of the bearing housing **20**. Each peak **44** is arcuately curved to bulge toward the outer circumferential surface of the top foil **30**. Each bump foil **40** has a corrugated shape in which the valleys **43** and the peaks **44** are alternately arranged in the circumferential direction B. Each bump foil **40** has a configuration in which the valleys **43** and the peaks **44** are arranged alternately from the fixed end **41** toward the free end **42**. The fixed end **41** and the free end **42** are each continuous with the peaks **44**.

As shown in FIG. **4**, the fixed end **41** of each bump foil **40** includes an insertion plate portion **41***a*. The insertion plate portion **41***a* extends in the axial direction A. The insertion plate portion **41***a* includes a first section **411***a*, a second section **412***a*, and bent plate portions **41***b*. The bent plate portions **41***b* are arranged at the opposite ends in the axial direction A of the insertion plate portion **41***a*. The portion including the first section **411***a* and the second section **412***a* is L-shaped when viewed in the axial direction A. The first section **411***a* is continuous with one of the peaks **44**. The first section **411***a* has a thickness in the radial direction C. The first section **411***a* extends in the axial direction A.

The second section **412***a* extends outward in the radial direction C from the edge of the first section **411***a* on the side opposite to the peak **44**. The second section **412***a* extends in the axial direction A. The second section **412***a* includes one of the short edges **40***b* of the bump foil **40**.

As shown in FIGS. **3** and **4**, the bent plate portions **41***b* are provided at the opposite ends in the axial direction A of the second section **412***a*. The second section **412***a* includes cutout portions **413***a* at the opposite ends in the axial direction A. The cutout portions **413***a* are formed in parts of the second section **412***a* that are in the vicinity of the first section **411***a*. The bent plate portions **41***b* are bent with respect to the second section **412***a* to extend in the circumferential direction B from the opposite ends in the axial direction A of the second section **412***a*. The bent plate portions **41***b* are formed integrally with the second section **412***a*. The bent plate portions **41***b* are bent with respect to a section of the insertion plate portion **41***a* other than the bent plate portions **41***b*. The bent plate portions **41***b* extend toward the same side in the circumferential direction B from the opposite ends of the insertion plate portion **41***a*. The bent plate portions **41***b* also extend in the radial direction C. The bent plate portions **41***b* may be fixed to the second section **412***a*, for example, by welding.

As shown in FIG. **4**, each bent plate portion **41***b* includes an end face **411***b* extending in the radial direction C. The end face **411***b* is flush with one of the long edges **40***a* of the bump foil **40**. That is, the length between the two end faces **411***b* of the bump foil **40** is equal to the length between the two long edges **40***a* of the bump foil **40**. The amount of extension of the end face **411***b* from the second section **412***a* in the circumferential direction B is at least greater than the thickness of the bump foil **40**. In a direction orthogonal to both the direction in which the end face **411***b* extends from the second section **412***a* and the thickness direction of the bent plate portion **41***b*, the length of the end face **411***b* is greater than the thickness of the bump foil **40**. The bent plate portion **41***b* has a size such that the bent plate portion **41***b* can be inserted into the holding groove **20***b*.

As shown in FIGS. **2** and **4**, the insertion plate portion **41***a* of each bump foil **40** is inserted into the corresponding holding groove **20***b*. The second section **412***a* and the bent plate portions **41***b* of each bump foil **40** are inserted into the corresponding holding groove **20***b*. The bump foil **40** includes the insertion plate portion **41***a*, which is inserted into the holding groove **20***b* and extends in the axial direction A. The insertion plate portion **41***a* of one of the three bump foils **40** is inserted into one of the holding grooves **20***b* together with the insertion plate portion **31***b* of the top foil **30**. The insertion plate portions **41***a* of the remaining two bump foils **40** are respectively inserted into the remaining two holding grooves **20***b*. As for the bump foils **40** adjacent to each other in the circumferential direction B, the free end **42** of one of the bump foils **40** is spaced apart from the fixed end **41** of the other bump foil **40** by a specified distance. The three bump foils **40** are disposed outward in the radial direction C of the top foil **30**. The three bump foils **40** elastically support the top foil **30** between the bearing housing **20** and the top foil **30**.

Retaining Member

As shown in FIG. **3**, each retaining member **50** has a circular and annular shape. The inner diameter of each retaining member **50** is the same as the outer diameter of the cylindrical surface **20***d* of the bearing housing **20**. In the axial direction A, one of the retaining members **50** is fitted in the cutout groove **20***c* formed in the first end portion **21** of the bearing housing **20**. In the axial direction A, the other retaining member **50** is fitted in the cutout groove **20***c* formed in the second end portion **22** of the bearing housing **20**. Each retaining member **50** contacts the corresponding annular surface **20***e*. Each retaining member **50** faces the corresponding holding grooves **20***b*, which open in the annular surface **20***e* in the axial direction A. The retaining members **50** are fixed to the bearing housing **20** by fixing means (not shown).

As shown in FIGS. **3** and **5**, each retaining member **50** faces the insertion plate portion **31***b* of the top foil **30** and the insertion plate portions **41***a* of the bump foils **40** in the axial direction A. Each retaining member **50** faces part of the bent plate portions **41***b* in the axial direction A. Each retaining members **50** is a retaining portion that faces the top foil **30** and the bump foils **40** in the axial direction A and prevents the top foil **30** and the bump foils **40** from being dislodged from the bearing housing **20**.

As shown in FIG. **6**, each retaining member **50** includes a facing surface **50**a that faces the insertion plate portions **31**b, **41**a in the axial direction A. The facing surface **50**a is a surface extending in the radial direction C. The facing surface **50**a extends in a direction orthogonal to the axis of the bearing housing **20**. In the axial direction A, a distance L**1** between the two facing surface **50**a is longer than a length L**2** between the end faces **411**b of the two bent plate portions **41**b. That is, in the axial direction A, the distance L**1** between the two facing surfaces **50**a is longer than the length L**2** of the insertion plate portion **41**a. The facing surfaces **50**a of the retaining members **50** are parallel to the end faces **411**b of the bent plate portions **41**b. The bent plate portions **41**b thus extend along the facing surfaces **50**a.

Operation of First Embodiment

Operation of the first embodiment will now be described.

As shown in FIG. **2**, when the rotary shaft **103** rotates in a rotation direction D and the rotation speed of the rotary shaft **103** reaches a specified rotation speed, an air film **105** is formed between the rotary shaft **103** and the top foil **30**. This separates the rotary shaft **103** from the top foil **30**. The rotary shaft **103** is supported by the air film **105** in a state of being separated from the top foil **30** in the radial direction Rd.

While the rotary shaft **103** is rotating, the rotary shaft **103** may be inclined with respect to the axis of the bearing housing **20**. If the rotary shaft **103** is inclined with respect to the axis of the bearing housing **20**, the thickness of the air film **105** is uneven in the axial direction A. In the axial direction A, dynamic pressure in the radial direction Rd generated by the air film **105** is low in some areas and high in other areas. Thus, at least one of the top foil **30** or the set of the bump foils **40** receives a moving force acting in the axial direction A. This causes at least one of the top foil **30** and the set of the bump foils **40** to be pressed against the retaining member **50** more than necessary.

Also, in the present embodiment, the thickness of the bump foils **40** is less than the thickness of the top foil **30**. If the bump foils **40** do not have the bent plate portions **41**b, and the thickness of the bump foils **40** is less than the thickness of the top foil **30**, the wear caused by contact between the bump foils **40** and the retaining members **50** may be greater than the wear caused by contact between the top foil **30** and the retaining members **50**.

In the present embodiment, the bent plate portions **41**b are in planar contact with the facing surfaces **50**a. This ensures a sufficient contact area between each bump foil **40** and the retaining members **50**. This reduces the contact pressure between the bump foils **40** and the retaining members **50**. Thus, even if the bump foils **40** are pressed against the retaining members **50** more than necessary, the wear caused by contact between the bump foils **40** and the retaining member **50** is reduced.

Advantages of First Embodiment

The present embodiment has the following advantages.

(1-1) The bent plate portions **41**b are in planar contact with the facing surfaces **50**a. This ensures sufficient contact area between the bump foils **40** and the retaining members **50**. This reduces the contact pressure between the bump foils **40** and the retaining members **50**. Thus, even if the bump foils **40** are pressed against the retaining members **50** more than necessary, the wear of the bump foils **40** and the retaining member **50** is reduced.

(1-2) If the bump foils **40** do not have the bent plate portions **41**b, and the thickness of the bump foils **40** is less than the thickness of the top foil **30**, the wear caused by contact between the bump foils **40** and the retaining members **50** may be greater than the wear caused by contact between the top foil **30** and the retaining members **50**.

In this regard, each bump foil **40** of the present embodiment includes the bent plate portions **41**b in the insertion plate portion **41**a. This further effectively reduces the wear caused by contact between the bump foils **40** and the retaining members **50**.

(1-3) The bent plate portions **41**b extend toward the same side in the circumferential direction B from the opposite ends of the insertion plate portion **41**a. Thus, as compared with a case in which the bent plate portions **41**b extend toward different sides from the insertion plate portion **41**a, the width of the holding groove **20**b in the circumferential direction B is reduced. Thus, the machining amount of the bearing housing **20** is reduced, which limits reduction in the strength of the foil bearing **10**.

Second Embodiment

A foil bearing according to a second embodiment will now be described with reference to FIGS. **7** and **8**. The same reference numerals are given to those components that are the same as the corresponding components of the first embodiment and detailed explanations are omitted.

Bent Plate Portion

As shown in FIG. **7**, an angle θ**1** formed between each bent plate portion **41**b and the second section **412**a is an obtuse angle. Each bent plate portion **41**b is connected to the second section **412**a to form an obtuse angle with the second section **412**a. The second section **412**a is a main plate portion that is located in the holding groove **20**b and extends in the axial direction A. The insertion plate portion **41**a includes the second section **412**a, which is a main plate portion located in the holding groove **20**b and extending in the axial direction A.

In the axial direction A, the distance L**1** between the two facing surface **50**a is longer than a length L**3** between the tips of the two bent plate portions **41**b. That is, in the axial direction A, the distance L**1** between the two facing surfaces **50**a is longer than the length L**3** of the insertion plate portion **41**a.

A bent portion **45** is formed at the boundary between the second section **412**a and each bent plate portion **41**b. The bent portion **45** has a spring structure. The angle θ**1** shown in FIG. **7** is an angle before the bump foil **40** and the retaining members **50** contact each other.

As shown in FIG. **8**, in the present embodiment, when the bent plate portion **41**b is pressed against the retaining member **50**, the angle θ**2** formed between the bent plate portion **41**b and the second section **412**a is smaller than the angle θ**1**. When the bump foil **40** is continuously pressed against the retaining member **50**, the bent plate portion **41**b continues to bend such that the area of the bent plate portion **41**b that contacts the facing surface **50**a of the retaining member **50** increases.

FIG. **8** shows a state in which part of the bent plate portion **41**b is in planar contact with the facing surface **50**a of the retaining member **50**. In this state, if the bump foil **40** is continuously pressed against the retaining member **50**, the bent plate portion **41***b* will eventually be deformed to extend along the facing surface **50***a* of the retaining member **50**, so that the angle θ**2** becomes 90 degrees. In the present embodiment, the material of the bump foils **40**, the thickness of the bump foils **40**, the rigidity of the bent plate portions **41***b*, and the angle θ**1** are adjusted such that the area of the bent plate portion **41***b* that contacts the facing surface **50***a* of the retaining member **50** increases as the bump foil **40** is pressed against the facing surface **50***a* of the retaining member **50**. Even if the bump foil **40** is continuously pressed against the retaining member **50**, the bent plate portion **41***b* does not necessarily need to be bent to a state of extending along the facing surface **50***a*.

Operation of Second Embodiment

Operation of the present embodiment will now be described.

In the present embodiment, the bent portion **45** between the bent plate portion **41***b* and the second section **412***a* has a spring structure. Thus, even if the bump foil **40** is pressed against the retaining member **50** more than necessary, the load applied to the bump foil **40** and the retaining member **50** is absorbed by the elasticity of the bent portion **45**. This reduces the contact pressure between the bump foil **40** and the retaining member **50** and thus reduces the wear caused by the contact between the bump foil **40** and the retaining member **50**.

Advantages of Second Embodiment

The present embodiment has the following advantages.

(2-1) The bent portion **45** between the bent plate portion **41***b* and the second section **412***a* has a spring structure. Thus, even if the bump foil **40** is pressed against the retaining member **50** more than necessary, the load applied to the bump foil **40** and the retaining member **50** is absorbed by the elasticity of the bent portion **45**. This reduces the contact pressure between the bump foil **40** and the retaining member **50** and thus reduces the wear caused by the contact between the bump foil **40** and the retaining member **50**.

(2-2) When the bump foil **40** is continuously pressed against the retaining members **50**, the bent plate portion **41***b* continues to bend such that the area of the bent plate portion **41***b* that contacts the facing surface **50***a* of the retaining member **50** increases. This ensures sufficient contact area between the bump foil **40** and the retaining member **50**. Thus, the contact pressure between the bump foil **40** and the retaining member **50** is reduced as the contact area between the bent plate portion **41***b* and the facing surface **50***a* increases. This further reduces the wear caused by the contact between the bump foil **40** and the retaining member **50**.

Third Embodiment

A foil bearing according to a third embodiment will now be described with reference to FIG. **9**. The same reference numerals are given to those components that are the same as the corresponding components of the first embodiment and detailed explanations are omitted.

Buffer Portion

As shown in FIG. **9**, the insertion plate portion **41***a* of each bump foil **40** includes buffer portions **41***c*. Each buffer portion **41***c* is a plate-shaped portion that is connected to the second section **412***a* and the bent plate portion **41***b*. The second section **412***a* is a main plate portion that is located in the holding groove **20***b* and extends in the axial direction A. The insertion plate portion **41***a* includes the bent plate portions **41***b*, the second section **412***a*, and the buffer portions **41***c*. An angle θ**3** formed between the buffer portion **41***c* and the second section **412***a* is an obtuse angle. An angle θ**4** formed between the bent plate portion **41***b* and the buffer portion **41***c* is an obtuse angle. The angles θ**3**, θ**4** shown in FIG. **9** are angles before the bump foil **40** and the retaining members **50** contact each other. A bent portion **47** is formed at the boundary between the second section **412***a* and the buffer portion **41***c*. The bent portion **47** has a spring structure. A bent portion **48** is formed at the boundary between the bent plate portion **41***b* and the buffer portion **41***c*. The bent portion **48** has a spring structure. In the same manner as the first embodiment, the bent plate portion **41***b* extends along the facing surface **50***a*.

In the axial direction A, the distance L**1** between the two facing surfaces **50***a* is longer than a length L**4** between the end faces **411***b* of the two bent plate portions **41***b*. That is, in the axial direction A, the distance L**1** between the two facing surfaces **50***a* is longer than the length L**4** of the insertion plate portion **41***a*. The buffer portion **41***c* is separated from the facing surface **50***a*. Therefore, a specified space S is formed between the buffer portion **41***c* and the facing surface **50***a*.

Operation of Third Embodiment

Operation of the present embodiment will now be described.

When the bump foil **40** is pressed against the retaining member **50** more than necessary, the end face **411***b* of the bent plate portion **41***b* comes into contact with the facing surface **50***a*. When the bent plate portion **41***b* is further pressed against the facing surface **50***a*, the angles θ**3**, θ**4** are increased. The buffer portion **41***c* is provided between the second section **412***a* and the bent plate portion **41***b* such that the angles θ**3**, θ**4** are changed due to contact between the facing surface **50***a* and the bent plate portion **41***b*.

Even if the bump foil **40** is pressed against the retaining member **50** more than necessary, the planar contact between the bent plate portion **41***b* and the facing surface **50***a* reduces the contact pressure between the bump foil **40** and the retaining member **50**.

Also, the bent portion **47** between the second section **412***a* and the buffer portion **41***c* and the bent portion **48** between the buffer portion **41***c* and the bent plate portion **41***b* each have a spring structure. Thus, even if the bump foil **40** is pressed against the retaining member **50** more than necessary, the load applied to the bump foil **40** and the retaining member **50** is absorbed by the elasticity of the two bent portions **47**, **48**. Thus, in addition to the planar contact between the bent plate portion **41***b* and the facing surface **50***a*, elastic deformation of the two bent portions **47**, **48** further reduces the wear caused by contact between the bump foil **40** and the retaining member **50**.

Advantage of Third Embodiment

The present embodiment has the following advantages.

(3-1) Even if the bump foil **40** is pressed against the retaining member **50** more than necessary, the planar contact between the bent plate portion **41***b* and the facing surface **50***a* reduces the contact pressure between the bump foil **40** and the retaining member **50**.

Also, the bent portion **47** between the second section **412***a* and the buffer portion **41***c* and the bent portion **48** between the buffer portion **41***c* and the bent plate portion **41***b* each have a spring structure. Thus, even if the bump foil **40** is pressed against the retaining member **50** more than necessary, the load applied to the bump foil **40** and the retaining member **50** is absorbed by the elasticity of the two bent portions **47**, **48**. Thus, in addition to the planar contact between the bent plate portion **41***b* and the facing surface **50***a*, elastic deformation of the two bent portions **47**, **48** further reduces the wear caused by contact between the bump foil **40** and the retaining member **50**.

Modifications

The above-described embodiments may be modified as follows. The above-described embodiments and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

The bent plate portions **41***b* may extend toward the opposite sides in the circumferential direction B from the opposite ends of the insertion plate portion **41***a*.

As shown in FIG. **10**, the bent plate portion **41***b* may be formed integrally with the first section **411***a* of the insertion plate portion **41***a*. The bent plate portion **41***b* are bent with respect to the first section **411***a* to extend in the radial direction C from the first section **411***a*. The amount of extension of the end face **411***b* from the first section **411***a* in the radial direction C is made at least greater than the thickness of the bump foil **40**. In a direction orthogonal to both the direction in which the end face **411***b* extends from the first section **411***a* and the thickness direction of the bent plate portion **41***b*, the length of the end face **411***b* is made greater than the thickness of the bump foil **40**. The bent plate portions **41***b* may be fixed to the first section **411***a*, for example, by welding.

As shown in FIG. **11**, the insertion plate portion **31***b* of the top foil **30** may include bent plate portions **31***c*, which extend along the facing surfaces **50***a* of the retaining members **50**. The insertion plate portion **31***b* may include bent plate portions **31***c*. In this modification, the bent plate portions **31***c* are provided at the opposite ends in the axial direction A of the insertion plate portion **31***b*. Each bent plate portion **31***c* includes an end face **311***c*. The end face **311***c* extends in parallel with the facing surface **50***a* of the retaining member **50**. In this modification, the bent plate portions **31***c*, **41***b* do not necessarily need to overlap with each other in the axial direction A.

In the axial direction A, the length between the end faces **311***c* of the two bent plate portions **31***c* may be equal to the length L**2**. That is, in the axial direction A, the distance L**1** between the two facing surfaces **50***a* is longer than the length L**2** of the insertion plate portion **31***b*. The length of the insertion plate portion **31***b* may be different from the length L**2**. If the distance L**1** is longer than the length of the insertion plate portion **31***b*, the length of the insertion plate portion **31***b* may be changed.

In the above-described modification shown in FIG. **11**, when the insertion plate portion **31***b* includes the bent plate portions **31***c*, a portion of the insertion plate portion **31***b* that is located in the holding groove **20***b* and extends in the axial direction A serves as the main plate portion of the insertion plate portion **31***b*. Each bent plate portion **31***c* may be connected to the main plate portion to form an obtuse angle with the main plate portion of the insertion plate portion **31***b*.

In the above-described modification shown in FIG. **11**, the insertion plate portion **31***b* may have the same configuration as the buffer portion **41***c* described in the third embodiment.

The bent plate portions **41***b* of the bump foil **40** may be omitted, and the top foil **30** may have the bent plate portions **31***c*. That is, it suffices if at least one of the insertion plate portion **31***b* of the top foil **30** or the insertion plate portion **41***a* of the bump foil **40** includes the bent plate portions **31***c*, **41***b*. If at least one of the bent plate portions **31***c*, **41***b* is provided, the wear caused by contact between the retaining member **50** and at least one of the top foil **30** and the bump foil **40** is reduced.

The thickness of the bump foil **40** may be the same as the thickness of the top foil **30**. The thickness of the top foil **30** may be smaller than that of the bump foil **40**.

The fixed end **31** of the top foil **30** does not necessarily need to have the cutout portion **31***a*.

When the two retaining members **50** are fixed to the bearing housing **20**, the bent plate portions **41***b* may already be in contact with the facing surfaces **50***a* of the retaining members **50**.

The positions of the cutout grooves **20***c* may be changed in the radial direction C if the retaining members **50** can be attached to the cutout grooves **20***c*.

The two retaining members **50** may be modified as follows. In the following description, the cutout grooves **20***c* are omitted from the bearing housing **20**.

As shown in FIG. **12**, each retaining member **50** includes a main body **51** and three fixing portions **52**. The main body **51** is circular and annular. The outer diameter of the main body **51** is slightly smaller than the inner diameter of the bearing housing **20**. The inner diameter of the main body **51** is larger than the outer diameter of the top foil **30**.

Each fixing portion **52** projects outward in the radial direction of the main body **51** from the outer circumferential surface of the main body **51**. The fixing portions **52** are arranged on the main body **51** at specified intervals in the circumferential direction of the main body **51**. The fixing portions **52** are arranged on the main body **51** to correspond to the holding grooves **20***b* of the bearing housing **20**.

The two retaining members **50** are arranged inside the bearing housing **20**. Each fixing portion **52** is arranged in the corresponding holding groove **20***b*. Each fixing portion **52** is press-fitted into the corresponding holding groove **20***b*. The retaining member **50** is thus fixed to the bearing housing **20**. The outer circumferential surface of the main body **51** extends along the inner circumferential surface **20***a* of the bearing housing **20**. The main body **51** faces part of the fixed end **31** of the top foil **30** and part of the fixed end **41** of the bump foil **40** in the axial direction A. The main body **51** faces the valleys **43** of the bump foil **40** in the axial direction A and faces part of each peak **44**. That is, the two retaining members **50** face the top foil **30** and the bump foils **40** in the axial direction A.

The fixing portions **52** of one of the retaining members **50** are press-fitted into the holding grooves **20***b* from the first end portion **21** of the bearing housing **20**. The fixing portions **52** of the other retaining member **50** are press-fitted into the holding grooves **20***b* from the second end portion **22** of the bearing housing **20**. The fixing portions **52** of the two retaining members **50** sandwich the insertion plate portions **41***a* of the bump foils **40** in the axial direction A. In one of the holding grooves **20***b*, the fixing portions **52** of the two retaining members **50** also sandwich the insertion plate portion **31***b* of the top foil **30** in the axial direction A. In this manner, the two retaining members **50** prevent the top foil **30** and the bump foils **40** from being dislodged from the bearing housing **20**. The fixing portions **52** preferably include facing surfaces **52***a* that face the insertion plate portions **31**b, **41**a in the axial direction A. The facing surfaces **52**a extend in the radial direction C.

As shown in FIG. **13**, for example, the insertion plate portion **41**a may include a bent plate portion **41**b only at one end in the axial direction A. Thus, the insertion plate portion **41**a only needs to include the bent plate portion **41**b at at least one of the opposite ends in the axial direction A. The retaining member **50** facing the other end of the insertion plate portion **41**a may include a retaining main body **55** and an elastic portion **56**. The retaining main body **55** is, for example, circular and annular. The elastic portion **56** is located at a portion of the retaining main body **55** that faces the insertion plate portion **41**a in the axial direction A. The elastic portion **56** is thus provided on the retaining main body **55**. The elastic portion **56** is, for example, a rubber member. The elastic portion **56** is elastically deformable. The elastic portion **56** may be, for example, a plastic coating layer formed by coating with plastic a portion of the retaining main body **55** that faces the insertion plate portion **41**a in the axial direction A. That is, the elastic portion **56** may have any configuration if it is provided on the retaining main body **55** and is elastically deformable. The surface of the elastic portion **56** opposite to the retaining main body **55** is a facing surface **50**a, which faces the insertion plate portion **41**a. The elastic portion **56** thus includes the facing surface **50**a.

In the above-described configuration, the elastic portion **56** includes the facing surface **50**a. Thus, for example, even if the end of the insertion plate portion **41**a that does not include the bent plate portion **41**b is pressed against the retaining member **50** more than necessary, the contact pressure between the bump foil **40** and the retaining member **50** is reduced by the elastic portion **56**. Thus, even if the bent plate portion **41**b is not provided at one of the opposite ends in the axial direction A of the insertion plate portion **41**a, the wear of the bump foil **40** and the retaining member **50** is reduced.

As shown in FIG. **14**, for example, the insertion plate portion **41**a may include a bent plate portion **41**b only at one of the opposite ends in the axial direction A. Thus, the insertion plate portion **41**a only needs to include the bent plate portion **41**b at at least one of the opposite ends in the axial direction A. One of the two retaining members **50** that faces the bent plate portion **41**b in the axial direction A may be made of a nonmagnetic material. The nonmagnetic material is, for example, stainless steel or aluminum. Further, the retaining member **50** that faces the end of the insertion plate portion **41**a that does not include the bent plate portion **41**b may be made of a magnetic material. The magnetic material is, for example, stainless steel or iron.

The retaining member **50** made of a magnetic material has a higher wear resistance than the retaining member **50** made of a nonmagnetic material. Thus, for example, even if the end of the insertion plate portion **41**a that does not include the bent plate portion **41**b is pressed against the retaining member **50** more than necessary, the wear of the retaining member **50** is reduced. Therefore, even if the bent plate portion **41**b is not provided at one of the opposite ends in the axial direction of the insertion plate portion **41**a, the wear of the retaining member **50** is reduced.

Taking wear resistance into consideration, it is preferable that the retaining member **50** be made of a magnetic material. If the retaining member **50** is made of a magnetic material, the coils of the electric motor **102** may generate heat due to the influence of the magnetic flux generated by the retaining member **50**. Thus, one of the two retaining members **50** that is closer to the electric motor **102** may be made of a nonmagnetic material. However, the wear resistance of the retaining member **50** made of a nonmagnetic material is lower than that of the retaining member **50** made of a magnetic material. Thus, the retaining member **50** made of a nonmagnetic material faces the bent plate portions **41**b in the axial direction A. Thus, even if the retaining member **50** close to the electric motor **102** is made of a nonmagnetic material, the contact pressure between the bump foils **40** and the retaining member **50** is reduced by planar contact between the bent plate portions **41**b and the facing surface **50**a. This reduces the wear caused by the contact between the bump foil **40** and the retaining member **50**.

In the modification shown in FIG. **12**, the main bodies **51** may be omitted, and only the fixing portions **52**, which are press-fitted into the holding grooves **20**b, may be used as retaining portions.

The cutout grooves **20**c of the bearing housing **20** may be omitted, and the two retaining members **50** may be fixed to the opposite end faces in the axial direction A of the bearing housing **20**. The retaining members **50** may cover the holding grooves **20**b in the axial direction A and include facing surfaces that face the insertion plate portions **31**b, **41**a. In this modification, the retaining members **50** may be fixed to the bearing housing **20** by, for example, bolts or adhesive. In short, the retaining members **50** may be fixed to the bearing housing **20** by any means. The retaining members **50** may be snap rings.

One of the two retaining members **50** may be omitted. In this case, the portions of the holding grooves **20**b that open in the first end portion **21** or the second end portion **22** of the bearing housing **20** are preferably closed. The opening of the holding grooves **20**b may be closed by a part of the bearing housing **20**. In this case, the portion of the bearing housing **20** that closes the opening of the holding grooves **20**b may be a retaining portion. In other words, the bearing housing **20** may include a retaining portion. The retaining portion of the bearing housing **20** includes a facing surface that faces the insertion plate portions **31**b, **41**a in the axial direction A.

The facing surfaces **50**a do not necessarily need to extend in a direction orthogonal to the axis of the bearing housing **20**. The facing surfaces **50**a may be inclined with respect to the axis of the bearing housing **20** if the facing surfaces **50**a extend in the radial direction C.

In the illustrated embodiment, three bump foils **40** are employed. However, a single substantially annular bump foil **40** may be employed. In this case, the bearing housing **20** may include only one holding groove **20**b.

The foil bearing **10** does not necessarily need to support the rotary shaft **103** of the centrifugal compressor **100** in the radial direction Rd. The apparatus in which the foil bearing **10** is employed may be changed.

Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

What is claimed is:

1. A foil bearing that supports a rotary shaft in a radial direction, the foil bearing comprising:
- a tubular bearing housing into which the rotary shaft is inserted;
- a top foil disposed between the rotary shaft and the bearing housing;
- a bump foil that is disposed between the bearing housing and the top foil to elastically support the top foil; and
- two retaining portions facing the top foil and the bump foil in an axial direction of the bearing housing and preventing the top foil and the bump foil from being dislodged from the bearing housing, wherein
- a holding groove is formed in an inner circumferential surface of the bearing housing to extend in the axial direction,
- the top foil and the bump foil each include an insertion plate portion, which is inserted into the holding groove and extends in the axial direction,
- each of the retaining portions includes a facing surface that faces the insertion plate portions in the axial direction,
- in the axial direction, a distance between the facing surfaces of the two retaining portions is longer than a length of the insertion plate portions, and
- at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil includes a bent plate portion at at least one of opposite ends in the axial direction, the bent plate portion being bent and contacting the corresponding facing surface during a rotation of the rotary shaft in a state in which the rotary shaft is inclined with respect to an axis of the bearing housing.

2. The foil bearing according to claim 1, wherein the bent plate portion extends along the corresponding facing surface.

3. The foil bearing according to claim 2, wherein
- the at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil includes:
  - the bent plate portion;
  - a main plate portion that is located in the holding groove and extends in the axial direction; and
  - a buffer portion that is a plate-shaped portion connected to the main plate portion and the bent plate portion, and
- the buffer portion is provided between the main plate portion and the bent plate portion such that contact between the facing surface and the bent plate portion changes an angle formed between the buffer portion and the main plate portion and an angle formed between the buffer portion and the bent plate portion.

4. The foil bearing according to claim 3, wherein a space is formed between the buffer portion and the facing surface.

5. The foil bearing according to claim 1, wherein
- each facing surface extends in a direction orthogonal to an axis of the bearing housing,
- each insertion plate portion includes a main plate portion that is located in the holding groove and extends in the axial direction, and
- the bent plate portion is connected to the corresponding main plate portion such that the bent plate portion and the corresponding main plate portion form an obtuse angle.

6. The foil bearing according to claim 1, wherein
- a thickness of the bump foil is less than a thickness of the top foil, and
- the bent plate portion is provided on the insertion plate portion of the bump foil.

7. The foil bearing according to claim 1, wherein
- each of the opposite ends of the at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil is provided with the bent plate portion, and
- the bent plate portions extend toward the same side in a circumferential direction of the bearing housing from the at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil.

8. The foil bearing according to claim 1, wherein
- only one of the opposite ends of the at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil is provided with the bent plate portion,
- the retaining portion that faces the other opposite end of the at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil includes a retaining main body and an elastic portion that is provided on the retaining main body and is elastically deformable, and
- the elastic portion forms the facing surface.

9. The foil bearing according to claim 1, wherein
- only one of the opposite ends of the at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil is provided with the bent plate portion,
- the retaining portion that faces the bent plate portion is made of a nonmagnetic material, and
- the retaining portion that faces the other opposite end of the at least one of the insertion plate portion of the top foil and the insertion plate portion of the bump foil is made of a magnetic material.

* * * * *